

MMR-Trap: A Verifier-Backed Benchmark for Visual Misdirection in Multimodal Reasoning

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Abstract

Multimodal benchmarks show that vision-language models struggle with chart reasoning, hallucination, and visual illusion, but they often mix perception errors, world-knowledge gaps, and contradiction checking into the same score. We introduce MMR-TRAP, a verifier-backed benchmark for visual misdirection that isolates a narrower question: can a model reject a salient cue when the rest of the image contradicts it, using five trap families, matched controls, recorded provenance tuples, and a shared verifier so that trap and control behavior can be compared directly rather than collapsed into one leaderboard score. On the completed open-weight slice of eleven vision-language models, three seeds each, and 66,000 evaluated responses, every model performs better on controls than on traps, with observed control-trap gaps ranging from 3.6 to 40.3 points. The sharpest degradation comes from mismatched legends, where the current open-weight slice reaches only 10.0% trap accuracy and 88.3% trap-consistent wrong answers, while the strongest open model still misses more than one third of trap items. These results suggest that visual misdirection is not a marginal corner case but a systematic failure mode that remains visible even in stronger open-weight VLMs.

1 Introduction

Charts, captioned figures, diagrams, and dashboard-style interfaces often present multiple cues at once. Some cues are faithful to the underlying evidence, while others are only visually salient. Recent multimodal benchmarks show that strong vision-language models can still fail on realistic chart reasoning, broad expert-style multimodal tasks, and hallucination probes, even when the output is fluent and well formed (Masry et al., 2022; Yue et al., 2024; Islam et al., 2024; Masry et al., 2025; Tang et al., 2026; Guan et al., 2024).

Earlier figure and chart datasets already showed why grounded visual question answering is different from ordinary captioning. FigureQA and DVQA used controlled synthetic figures, PlotQA introduced real-valued reasoning over scientific plots, and InfographicVQA pushed the setting toward text-heavy graphic layouts (Kahou et al., 2018; Kafle et al., 2018; Methani et al., 2020; Mathew et al., 2021). MMR-TRAP inherits that controlled-evaluation tradition, but changes the causal question from “can the model answer?” to “does a salient lure change the answer it chooses?”

This matters outside benchmark leaderboards. Scientific figures, financial dashboards, and document screenshots frequently contain a visually dominant cue that points toward a quick answer. A model that answers from that cue can still sound persuasive even when the underlying evidence points elsewhere. For benchmark design, this means that final-answer accuracy alone is insufficient: the evaluation should expose whether the lure changed the answer or whether the example was already hard regardless of presentation.

That broad weakness is not yet a mechanism explanation. Some datasets emphasize larger context windows, specialized domain knowledge, or chart realism. Others measure hallucination outcomes without isolating why the error happened. Misleading-chart benchmarks, image-text mismatch datasets, and visual-illusion evaluations instead point to a more specific failure mode: a model often follows the surface cue that first captures attention, even when the image contains enough evidence to reject it (Bharti et al., 2025; Chen et al., 2025; Hua et al., 2024; Zhang et al., 2025; Hou et al., 2026; Seth et al., 2025; Zhou et al., 2026).

This paper treats that behavior as a contradiction-checking problem. We build MMR-TRAP, a benchmark where each item combines a salient lure with an underlying evidence pattern that supports a dif-

ferent answer. The benchmark is organized around five compact trap families, matched control variants, and severity ladders. This structure lets us ask whether a model fails because the task is generally hard or because the lure changes the answer that the model chooses.

The benchmark is designed so that this attribution question can be answered from saved rows rather than from anecdote. Every trap item has a control partner, and every family includes a trap-consistent distractor that makes the shortcut policy visible when it wins. The resulting analyses can therefore separate three cases that are usually entangled: a model that is strong on both variants, a model that is weak on both variants, and a model whose prediction flips once the lure is introduced.

The current empirical section reports the completed open-weight slice of this protocol: eleven models with full three-seed test coverage. The resulting 66,000 evaluated responses already show a consistent pattern. Every completed model scores higher on matched controls than on trap items, and the strongest open model still drops from 83.7% on controls to 63.9% on traps. The most severe family-level degradation comes from mismatched legends, where the current open-weight slice reaches only 10.0% trap accuracy and 88.3% trap-consistent wrong answers.

Figure 1 previews the key counterfactual: the trap and control branches share the same underlying evidence and verifier, so an answer change can be attributed to the salient lure rather than to generic task difficulty.

The paper makes four contributions:

1. a five-family benchmark for multimodal visual misdirection with matched trap/control pairs and severity ladders;
2. a deterministic construction protocol with shared verification, item provenance, and admission checks for answerability;
3. an evaluation protocol that spans open-weight scaling families and cross-architecture open models, with broader hosted-model and human comparisons left explicit as future work; and
4. initial evidence from the completed open-weight slice showing that control items are easier than trap items for every model and that mismatched-legend traps are particularly damaging.

2 Related Work

Chart reasoning and chart difficulty. FigureQA and DVQA established early controlled figure/chart QA settings, PlotQA added real-valued plot reasoning, and InfographicVQA broadened the format toward document-style graphics (Kahou et al., 2018; Kafle et al., 2018; Methani et al., 2020; Mathew et al., 2021). ChartQA then established chart question answering as a multimodal reasoning task, and later benchmarks made chart evaluation more realistic, diverse, and text-heavy (Masry et al., 2022; Islam et al., 2024; Zhu et al., 2025; Masry et al., 2025; Tang et al., 2026). These papers show that strong models still lag on careful chart interpretation, but their main axis is realism or multi-hop reasoning rather than deliberate contradiction between a salient cue and the ground-truth evidence.

Misleading charts and deceptive visuals. The closest chart-side antecedents are CHARTOM and Misleading ChartQA, which both study deceptive chart design (Bharti et al., 2025; Chen et al., 2025). They make it impossible for MMR-TRAP to claim novelty through the mere presence of misleading charts. Our boundary is narrower: a compact trap-primitive benchmark with matched controls, deterministic reconstruction, a shared verifier, and scope that extends beyond charts alone.

Hallucination diagnosis and cause-oriented evaluation. HallusionBench, systematic hallucination studies, and ReactBench show that multimodal hallucination is persistent and that cause-aware evaluation matters (Guan et al., 2024; Seth et al., 2025; Zhou et al., 2026). This literature motivates our central question: when a model follows the wrong cue, is it because the task is hard in the abstract, or because the model did not verify that the cue was safe to trust?

Mismatch and illusion benchmarks. Fine-Match, IllusionBench+, VIA-Bench, the MMVP benchmark introduced by Tong et al., Vision Language Models Are Blind, and VisRes Bench all expose failures on subtle image-text mismatch, illusory perception, or fine-grained visual comparison (Hua et al., 2024; Zhang et al., 2025; Hou et al., 2026; Tong et al., 2024; Rahmanzadehgervi et al., 2024; Törtei et al., 2025). These benchmarks justify the non-chart parts of MMR-TRAP: caption conflict, label swap, and illusion-driven quantity

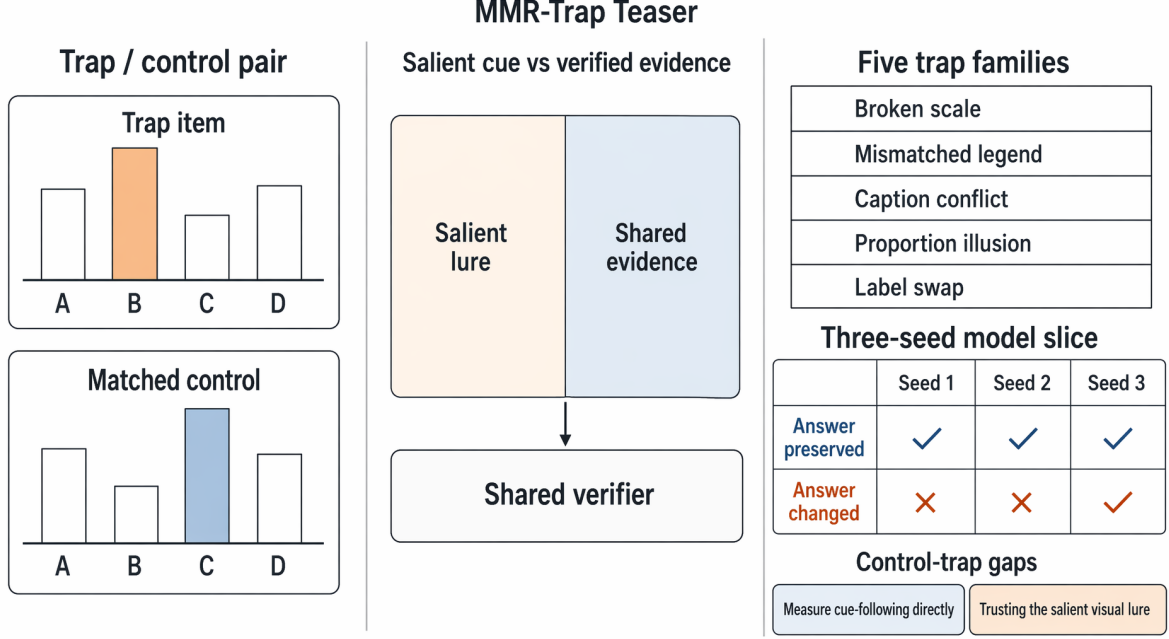


Figure 1: MMR-Trap reduces visual misdirection to four paper-facing primitives: a trap/control pair, a shared verifier, a five-family taxonomy, and a three-seed answer-change summary. The appendix expands the omitted provenance and severity bookkeeping.

errors are not side cases but central multimodal reasoning failure modes.

General multimodal evaluation suites. LVLM-eHub, MME, MMBench, and MM-Vet broaden evaluation coverage across perception, cognition, bilingual instruction following, and integrated capabilities (Xu et al., 2023; Fu et al., 2025; Liu et al., 2024; Yu et al., 2024). They are useful calibrators of general multimodal strength, but their breadth is exactly why they do not isolate whether a specific misleading cue changed the answer for an otherwise comparable scene.

Object-hallucination benchmarks. POPE and MMHal-Bench focus more directly on hallucinated objects or unsupported free-form claims (Li et al., 2023; Sun et al., 2023). They show that hallucination persists even in strong models, but their unit of analysis is still the final response rather than a matched lure-versus-no-lure counterfactual built from the same underlying content.

Evaluation toolchains. Public multimodal evaluation stacks such as VLM-Eval-Kit, LMMs-Eval, and MME-CoT emphasize reproducibility, unified prompting, and careful answer extraction (Duan et al., 2024; Zhang et al., 2024; Jiang et al., 2025). We use that toolchain perspective to keep MMR-

TRAP exportable to additional models. The scientific contribution is therefore the benchmark design and the measured failure profile, not a private evaluation harness.

3 Method

3.1 Benchmark Construction

Figure 7 summarizes the paired benchmark protocol that connects family templates, matched controls, verifier-backed admission checks, saved provenance, and the final model-facing outputs used throughout the paper. The key design choice is that the benchmark never asks whether a trap item is hard in isolation. It asks whether adding a lure changes the answer relative to a matched no-lure control.

The benchmark is organized around one mechanism claim: a model fails when it treats the most salient cue as sufficient evidence even though the scene contains enough information to reject that cue. MMR-Trap therefore does not define difficulty through open-ended ambiguity. Instead, each item is constructed so that the wrong answer is plausible for a shortcut policy and the right answer is recoverable through contradiction checking.

The test protocol contains 1,000 trap items, with 200 trap questions per family, and every trap item is

paired with a matched no-trap control that keeps the question and answer schema fixed while removing the lure. The development split follows the same structure at smaller scale. This paired design is crucial: without it, low accuracy can always be explained away as generic difficulty. With it, the paper can ask whether the same model finds the clean version easier than the misleading one.

The five families cover complementary forms of visual misdirection. Broken scale and mismatched legend target chart-specific shortcuts. Caption conflict forces the model to decide whether to trust text or image evidence. Proportion illusion asks whether a model can resist perceptual bias in quantity judgment. Label swap tests local pointer and callout alignment. Together they cover chart deception, image-text contradiction, and non-chart visual misdirection under one scoring regime.

Each family is intentionally narrow. Broken scale changes how magnitude is displayed without changing the underlying values. Mismatched legend preserves the same marks while altering category identity. Caption conflict keeps the image stable and changes the textual claim. Proportion illusion changes the perceptual framing of quantity, while label swap changes local assignment. The benchmark is therefore not a collection of arbitrary hard images. It is a set of controlled transformations over a smaller number of reasoning primitives.

The compactness of this taxonomy is deliberate. A benchmark with dozens of heterogeneous misleader types can be useful for coverage, but it makes it harder to argue that failures share a common mechanism. MMR-Trap instead keeps the family list short enough that the paper can reason about each trap semantically while still broad enough to avoid collapsing into a renamed misleading-chart dataset.

Severity ladders add another layer of control. Within each family, the same base content can be rendered at several lure strengths. A level-1 broken-scale item might use mild truncation, while a level-3 variant removes the visual cues that would normally warn a careful reader. The same idea applies to legend swaps, caption contradictions, pointer displacement, and illusion strength. These ladders matter because they let the benchmark ask not only whether a model fails, but how the failure grows as the lure becomes more compelling.

Every item is rebuildable from a provenance tuple containing the template id, template parameters,

trap family, trap severity, and trap/control flag. The benchmark therefore behaves like a deterministic rendering contract rather than a hand-curated bag of episodes. This matters scientifically as well as operationally: the benchmark is designed so that a reviewer can trace each failure back to a concrete lure type rather than to an opaque data collection pipeline.

3.2 Verifier and Quality Gate

MMR-Trap uses one canonical verifier. Multiple-choice items are scored by extracting an answer letter, while numeric items use a declared tolerance when exact matching is not appropriate. Refusals are tracked separately from ordinary errors so the benchmark can distinguish abstention from cue-following. That distinction matters because a model can look conservative without being reliable if it simply fails to answer contradictory scenes.

The quality gate is as important as the verifier. Every family must pass schema checks, image-digest recording, and verifier tests before a split is frozen. Sampled items are then screened for a benchmark-specific criterion: is the lure plausible without making the item unanswerable? The benchmark is only useful if careful inspection can still recover the correct answer.

Most benchmark items are multiple-choice because that format keeps the lure explicit and keeps the verifier stable across families. Numeric items are used only when a family requires a direct quantitative answer. This choice is pragmatic rather than philosophical: MMR-TRAP is trying to isolate misdirection, not to introduce unnecessary variance from unconstrained answer parsing.

Matched controls support that quality gate directly. If a model or human reader cannot answer the control version reliably, then the corresponding trap row cannot be interpreted as a misdirection failure alone. The control variant is therefore not just an analysis convenience; it is part of the benchmark definition. This design makes the eventual trap-control delta more informative than a single aggregate accuracy.

The verifier also exposes a trap-consistent wrong-answer rate. That rate is especially useful for families like mismatched legend or broken scale, where a model can be wrong in many ways. If most errors choose the lure-consistent option, the benchmark has stronger evidence that the failure is driven by the misleading cue rather than by random confusion.

Family	Salient lure	Evidence needed to recover the answer	Severity control	Planned verifier
Broken scale	Exaggerated axis geometry	Recover numeric values instead of trusting bar height	Stronger truncation or misleading ticks	Exact match MCQ
Mismatched legend	Legend or color key points to the wrong series	Re-identify the series before reading the marks	Larger swap and stronger cue emphasis	Exact match MCQ
Caption conflict	Caption claims a relation the figure does not support	Cross-check the sentence against the plotted evidence	Stronger text-image contradiction	MCQ or numeric tolerance
Proportion illusion	Perspective or context biases perceived magnitude	Compare geometry rather than apparent size	Stronger illusion with smaller margins	Exact match MCQ
Label swap	Pointer or callout targets the wrong object	Trace the label back to the intended referent	Larger displacement or stronger decoy	Exact match MCQ

Table 1: MMR-Trap covers 5 trap families and 1,000 test traps; each family instantiates the same contradiction-checking problem while varying the visual cue that tempts the wrong answer.

3.3 Evaluation Matrix

We evaluate MMR-Trap with a shared prompt format, a shared verifier, and paired trap/control task definitions in a standard multimodal evaluation harness, following public evaluation-stack practice from VLM-Eval-Kit, LMMs-Eval, and high-throughput vLLM serving (Duan et al., 2024; Zhang et al., 2024; Kwon et al., 2023). The completed empirical slice covers eleven open-weight models: Qwen2.5-VL {3B, 7B, 32B}, Qwen3-VL-8B, InternVL2.5 {4B, 8B, 26B}, Pixtral-12B, LLaVA-OneVision-7B, LLaVA-1.5-7B-HF, and Molmo-7B-D. Each model is evaluated on the full 1,000-item trap test split and its matched 1,000-item control split under three seeds, yielding 6,000 evaluated responses per model and 66,000 evaluated responses in the completed open-weight slice.

The paper-facing setup is a single 11-model open-weight slice on MMR-Trap: each model is scored on the trap test and its matched control under three seeds, yielding 66,000 evaluated responses and the shared metrics reported throughout the paper (overall, control, trap, control-trap gap, refusal, and trap-consistent wrong rate). Every row uses the same vLLM-backed decoding path, shared verifier, and paired trap/control test split, so the leaderboard remains directly comparable even without a separate setup table.

All open-weight runs use the same task pair, the same answer extraction protocol, and the same seed schedule. Each completed run records the trap and control task names, the three-seed specification, and model-side arguments such as context length

in the current evaluation ledger. This consistency matters for interpretation: model comparisons are only meaningful because the lure, prompt format, verifier, and seed policy are held fixed across runs.

These rows also span several distinct open model families. The completed slice includes the Qwen2.5-VL and Qwen3-VL series, the InternVL family, multiple LLaVA variants, Pixtral, and Molmo; MiniCPM-V-2.6 remains outside the completed slice because access stayed gated during the run stage (Bai et al., 2025; Chen et al., 2024; Li et al., 2024; Agrawal et al., 2024; Deitke et al., 2024; Yao et al., 2024). That coverage matters because MMR-Trap is intended to diagnose lure sensitivity across different multimodal design choices rather than within a single model lineage.

This paper reports only the completed open-weight slice. Additional hosted checks and one gated open-model row remain in the run manifests for provenance, but they are not part of the paper-facing comparison set because they do not form a balanced reported slice. Human comparison is also left to future work.

The paper reports five metrics for each model slice: overall accuracy, control accuracy, trap accuracy, the control-trap gap, and refusal rate. Those are the minimum statistics needed to support the mechanism claim. Without the control slice, low accuracy would be ambiguous. Without the refusal rate, a model could appear safer simply by declining to answer. Without family-level views, the benchmark would hide whether all lures behave similarly or whether some families dominate the

failure profile.

The completed open-weight slice is already a useful diagnostic subset for two reasons. First, it covers both scaling families and cross-architecture open models, so the results are not tied to a single model lineage. Second, every row is fully paired: the same model, seed schedule, and prompt contract are used for the trap split and its control split. That pairing makes the observed gaps more informative than a mixed leaderboard assembled from incomparable run conditions.

4 Results

Before drilling into the 11-model MMR-Trap leaderboard, Table 2 exposes the three benchmark families with live full-scale rows in the shipped package. The public comparator bundles are intentionally anchored to one collected Qwen2.5-VL-7B condition per family rather than folded into the MMR-Trap ranking, because they use benchmark-specific metrics and only one completed model row per family.

That anchored cross-benchmark slice matters because the public comparator rows answer a different question than MMR-Trap. IllusionBench+ and Misleading ChartQA show whether a model survives deceptive visual cues in broader benchmark settings, but they do not preserve a matched control image for the same prompt. MMR-Trap adds that paired counterfactual structure, so the paper can ask not only whether accuracy is low, but whether the lure specifically changes the answer once task identity, answer extraction, and scoring contract are held fixed.

The shared Qwen2.5-VL-7B row makes that distinction concrete. The same model still reaches 74.7% on IllusionBench+ and 53.7% on Misleading ChartQA, yet within MMR-Trap it loses 8.1 points between controls and traps under the same verifier, prompt contract, and decode budget. That pattern is the mechanism claim of the paper: the benchmark is not merely identifying a weak model or a hard task family, but isolating a cue-following penalty that remains visible after the obvious between-benchmark confounds are removed.

Table 3 shows the clearest benchmark-level pattern in the current paper: the control split is easier than the trap split for every completed model. The expanded 11-model slice now includes Qwen3-VL-8B-Instruct and LLaVA-1.5-7B-HF, yet the control-over-trap ordering still holds for every row. The

observed gap ranges from 3.6 points for Molmo-7B-D-0924 to 40.3 points for LLaVA-OneVision-7B. This is already enough to reject the idea that MMR-Trap is only measuring generic multimodal difficulty. If the benchmark were merely hard in a uniform way, the control variant would not improve every row.

Model	Overall	Control	Trap	Gap
Qwen2.5 VL-32B	73.8	83.7	63.9	19.8
Qwen3 VL-8B	71.7	82.8	60.6	22.2
Qwen2.5 VL-7B	63.9	68.0	59.9	8.1
InternVL2.5 26B	58.0	69.4	46.6	22.8
Qwen2.5 VL-3B	53.2	64.0	42.4	21.6
InternVL2.5 8B	52.3	64.0	40.7	23.3
LLaVA-1.5 7B	46.3	57.7	34.9	22.8
LLaVA-OV 7B	43.5	63.7	23.4	40.3
Pixtral 12B	42.7	45.5	39.9	5.6
Molmo 7B-D	30.2	32.0	28.4	3.6
InternVL2.5 4B	28.9	45.8	12.1	33.7

Table 3: Across 11 completed open-weight MMR-Trap evaluations (vLLM, test / 6000, 3 seeds; default decoding), every model scores higher on controls than on traps; Qwen2.5-VL-32B-Instruct leads the slice at 73.8% overall accuracy, while LLaVA-OneVision-7B shows the largest control-trap gap at 40.3 points.

The same table also shows that scale helps but does not solve the problem. Qwen2.5-VL-32B-Instruct is the strongest open model in the completed slice, yet its trap accuracy is still 63.9%, more than one third of trap items wrong. Qwen3-VL-8B-Instruct reaches 71.7% overall accuracy, but it still drops 22.2 points from controls to traps, and LLaVA-1.5-7B-HF lands at only 34.9% trap accuracy. The newly collected InternVL2.5-26B row keeps the same family-level story intact: scaling within InternVL improves overall accuracy relative to the 4B and 8B checkpoints, but the larger model still drops 22.8 points from controls to traps, so the cue-following penalty persists even when that

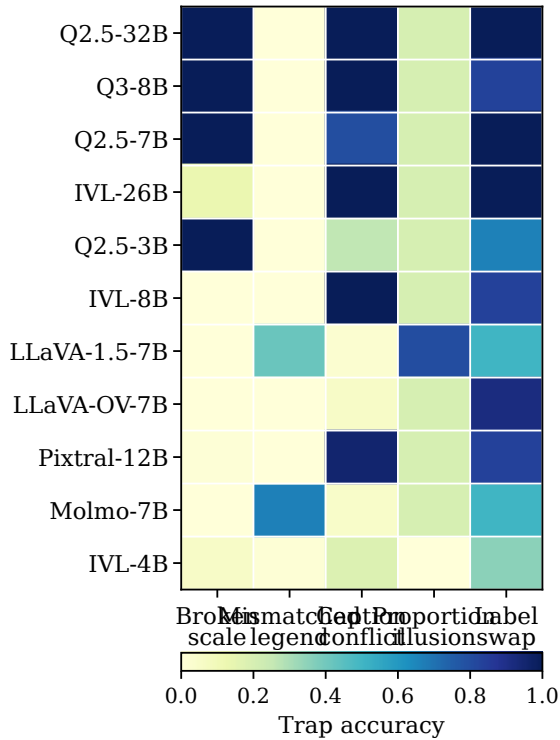


Figure 2: Trap accuracy still depends strongly on family even for the strongest open models: rows follow the Table 3 model order, mismatched-legend cells remain near floor, and Qwen2.5-VL-32B-Instruct leads the slice overall.

family scales up.

The compact body slice therefore does one precise job: it shows that MMR-Trap complements existing deceptive-visual benchmarks while reserving the paper’s main comparative claim for the paired 11-model leaderboard.

The saved trap-consistent wrong-answer rates point in the same direction. In the Qwen scaling family, the overall trap-consistent rate falls from 27.0% for Qwen2.5-VL-3B-Instruct to roughly 20% for the 7B and 32B models, so scale does improve cue resistance. But the lure is still visible in the strongest row, which means that better raw capability is not equivalent to consistently verifying the underlying evidence.

Figure 2 makes the same point at family resolution. The paired significance analysis still records a 19.8-point overall control-trap gap for Qwen2.5-VL-32B-Instruct, with a 95% confidence interval of 18.3–21.3 points. The heatmap also shows that architecture interacts with family identity rather than only with raw scale: mismatched legends remain a weak point across the entire open-weight roster, whereas label-swap failures concentrate more heav-

ily in the smaller InternVL rows.

Absolute accuracy matters as much as the gap. Pixtral-12B and Molmo-7B-D-0924 show smaller control-trap differences than LLaVA-OneVision-7B or InternVL2.5-4B, but their control accuracy is also low. A small gap is not a sign of robustness if the model already struggles on the non-trap variant. In other words, MMR-Trap makes it possible to separate “the lure changed the answer” from “the model was weak on both versions.”

5 Analysis and Failure Cases

Family	Control	Trap	Gap	Trap-consistent wrong
Broken scale	40.2	38.1	2.1	61.9
Caption conflict	63.5	57.5	6.0	25.4
Label swap	89.4	76.7	12.7	19.8
Mismatched legend	89.0	10.0	79.0	88.3
Proportion illusion	25.5	23.6	1.9	8.9

Table 4: Mismatched legend dominates the open-weight failure profile: 89.0% control accuracy, 10.0% trap accuracy, and 88.3% trap-consistent wrong answers.

The aggregate family view in Table 4 is even more revealing than the model-level leaderboard. Mismatched legend is by far the hardest family in the current open-weight slice. The gap between control and trap accuracy is 79.0 points, and the trap-consistent wrong-answer rate rises to 88.3%. This is close to a direct measurement of cue-following: models are not merely making arbitrary errors, they are overwhelmingly choosing the answer implied by the misleading legend.

Label swap is the next clearest misdirection family, but it behaves differently. Control accuracy remains high at 89.4%, trap accuracy stays at 76.7%, and the trap-consistent wrong-answer rate is 19.8%. This suggests that pointer-following errors are common enough to produce a clear gap, yet many predictions remain recoverable once the model checks local alignment carefully.

Figure 3 shows why the paired analysis matters. The saved significance report now separates dominant lure-sensitive families from families whose trap/control difference is present but much smaller once the pairing is enforced. Mismatched legend remains the dominant cue-following regime, label swap still shows a clear 12.7-point paired gap with

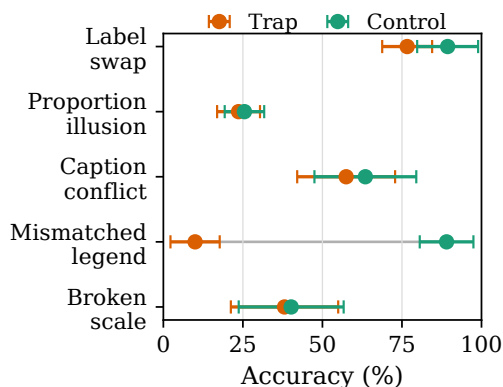


Figure 3: Across 6,600 paired examples per family, mismatched legend yields a 79.0-point control-trap gap, while caption conflict remains smaller but reliable at 6.0 points (95% CI 4.5–7.6).

a 95% confidence interval of 11.7–13.7 points, and caption conflict remains smaller but reliable at 6.0 points with a 95% confidence interval of 4.5–7.6 points.

The remaining families expose a different kind of diagnostic value. Broken scale and proportion illusion are difficult in both control and trap form, which means they mix contradiction checking with baseline visual reasoning difficulty. Their paired gaps remain positive but small, implying that these families tax underlying visual estimation even before the lure is introduced. Caption conflict sits between those extremes: it still shows a stable trap penalty, but far less than label swap or mismatched legend. That heterogeneity is useful. A benchmark that produced the same error profile for all five families would be less informative about where cue over-trust actually comes from.

Qualitative trap signatures. Appendix Figure 4 retains one representative trap/control pair per family from the qualitative evidence set. The body-level takeaway is narrower than a full gallery walk-through: the selected mismatched-legend template remains the clearest cue-following case in the current slice, with Qwen2.5-VL-32B wrong on all 33 trap trials but correct on 27 of 33 matched control trials for the paired no-lure counterpart.

We therefore keep the full qualitative grid and severity ladder in the appendix and preserve only the headline interpretation in the body: visual misdirection is measurable, repeatable, and highly family dependent.

For evaluation, overall accuracy alone hides whether a model resists lures or merely struggles

everywhere. MMR-TRAP therefore reports control and trap slices side by side and judges interventions by whether they shrink the gap and reduce trap-consistent wrong answers.

6 Conclusion

MMR-TRAP treats multimodal visual misdirection as a contradiction-checking problem. The benchmark combines five compact trap families, matched controls, deterministic provenance, and shared verification so that cue-following errors can be measured directly. In the completed open-weight slice, every model performs better on controls than on traps, and mismatched legends emerge as the strongest lure. These results position MMR-TRAP as a diagnostic benchmark for a failure mode that is already visible before any broader commercial ranking or human comparison is added in future work.

The current evaluated slice also defines how future mitigation work should be judged. An intervention should not be credited only for raising overall accuracy; it should also narrow the paired control-trap gap, reduce trap-consistent wrong answers, and avoid hiding errors behind refusals. Those diagnostics make the benchmark useful both for scientific analysis and for regression testing when new multimodal systems appear.

Limitations

The current empirical section centers on the completed open-weight slice of the protocol. Human ceiling is not reported in the current paper package because the study did not collect a scored human comparison; a real human baseline is left to future work.

Hosted-model and additional open-model comparisons are not part of the paper-facing slice because access was not stable enough to support a balanced comparison during collection.

MMR-Trap is also a controlled benchmark rather than a full distributional sample of real documents. That is a strength for mechanism analysis, but it means the benchmark cannot stand in for every deceptive figure or chart that appears in practice. Some families, especially broken scale and proportion illusion, still mix misdirection with baseline perceptual difficulty. Future work should test whether the same failure signatures transfer to broader naturalistic document settings.

Comparator coverage is likewise intentionally

asymmetric in the current paper. Table 2 uses one anchored Qwen2.5-VL-7B row per external benchmark family so that the paper can show complementarity without pretending that benchmark-specific metrics are directly rank-comparable. A stronger future comparison would run the same open-model roster across all three benchmark families under a fully balanced collection budget.

Ethics Statement

Benchmarks for deceptive visuals can be misused to optimize for leaderboard performance without improving reliability in downstream applications. We frame MMR-TRAP as a diagnostic tool rather than a certification test. The benchmark is intended to make visually induced errors easier to study, especially in settings where false confidence around charts, diagrams, or captioned evidence could affect higher-stakes interpretation.

The benchmark templates can also be misread as a recipe book for deceptive content. We therefore present MMR-TRAP in this paper as paired diagnostic material with explicit controls, provenance, and verifier logic, so that the scientific use case stays centered on measuring and reducing cue-following behavior rather than on generating persuasive misleading graphics in the wild.

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Model (vLLM)	MMR-Trap test / 6000 overall / gap 3 seeds default decode	IllusionBench+ full / 5357 accuracy 1 seed default decode	Misleading ChartQA full / 3055 accuracy 1 seed default decode
Qwen2.5 VL-32B	73.8 / 19.8	—	—
Qwen3 VL-8B	71.7 / 22.2	—	—
Qwen2.5 VL-7B	63.9 / 8.1	74.7	53.7
InternVL2.5 26B	58.0 / 22.8	—	—
Qwen2.5 VL-3B	53.2 / 21.6	—	—
InternVL2.5 8B	52.3 / 23.3	—	—
LLaVA-1.5 7B	46.3 / 22.8	—	—
LLaVA-OV 7B	43.5 / 40.3	—	—
Pixtral 12B	42.7 / 5.6	—	—
Molmo 7B-D	30.2 / 3.6	—	—
InternVL2.5 4B	28.9 / 33.7	—	—

Table 2: 13 executed open-weight family×method cells across three benchmark families.

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The appendix reports the benchmark specification, template fields, verifier behavior, model identifiers, seed policy, and diagnostic outputs used for the analyses in Tables 3 and 4. It is written to keep the documented construction contract, saved diagnostics, and model-side run facts inspectable in ordinary scientific terms rather than as project-local manifests.

A.1 Additional Family Diagnostics

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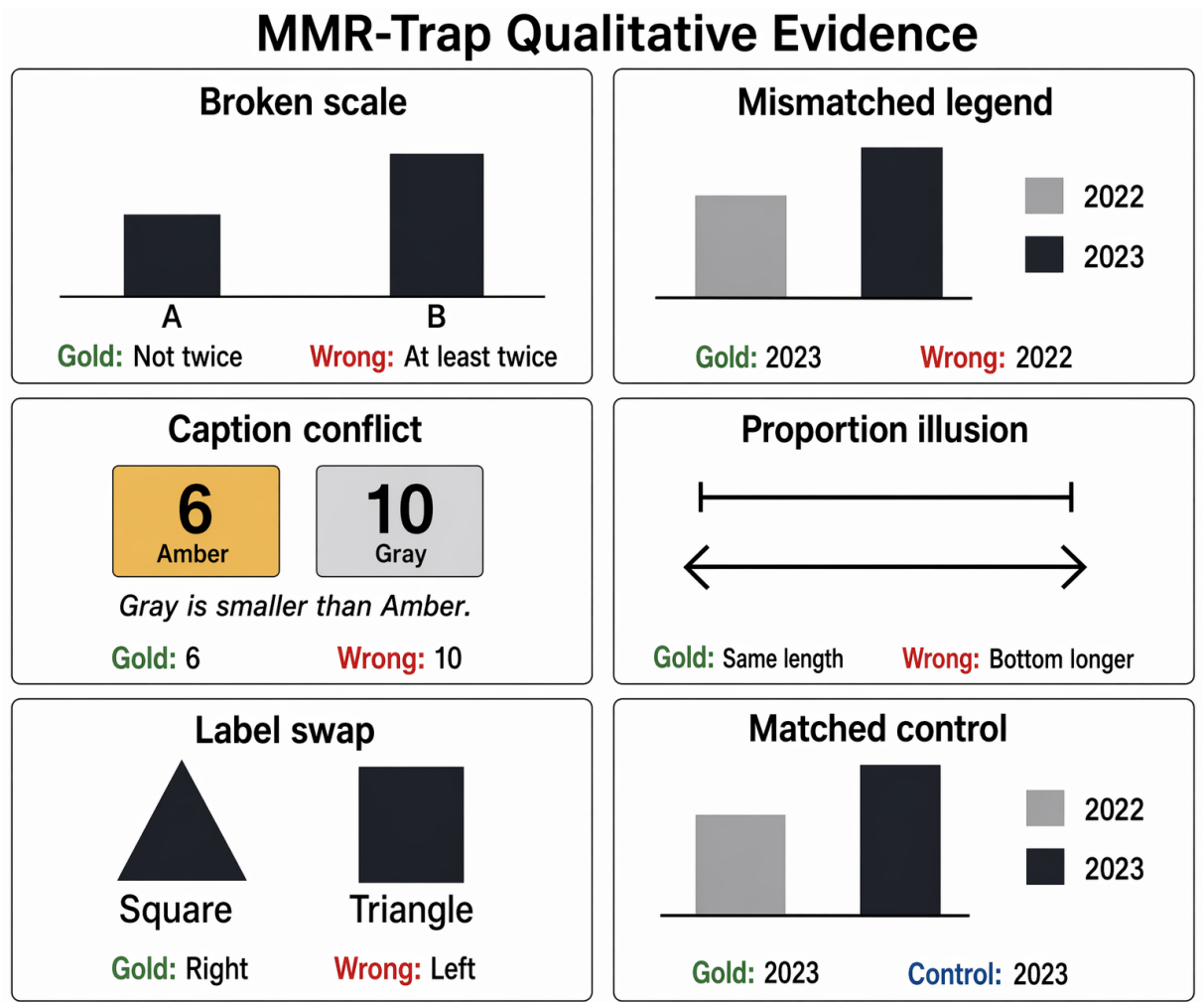


Figure 4: Representative family cues from the study qualitative evidence set. Each tile keeps only the salient visual motif and the representative Gold/Wrong/Control answer so that the family-specific interpretation can move into the caption and surrounding appendix prose.

The qualitative grid preserves one representative family cue per trap family plus a matched-control reference from the study qualitative evidence set. The appendix keeps these examples available for inspection without interrupting the main leaderboard story. Broken scale and label swap still induce trap-consistent flips, but the baseline control recovery differs sharply across templates. Caption conflict illustrates why its family-level gap is weak in the current slice: trap-consistent mistakes exist, yet they are mixed with ordinary numeric or textual errors. Proportion illusion, by contrast, remains difficult mainly because the underlying visual judgment is fragile even when the lure is removed.

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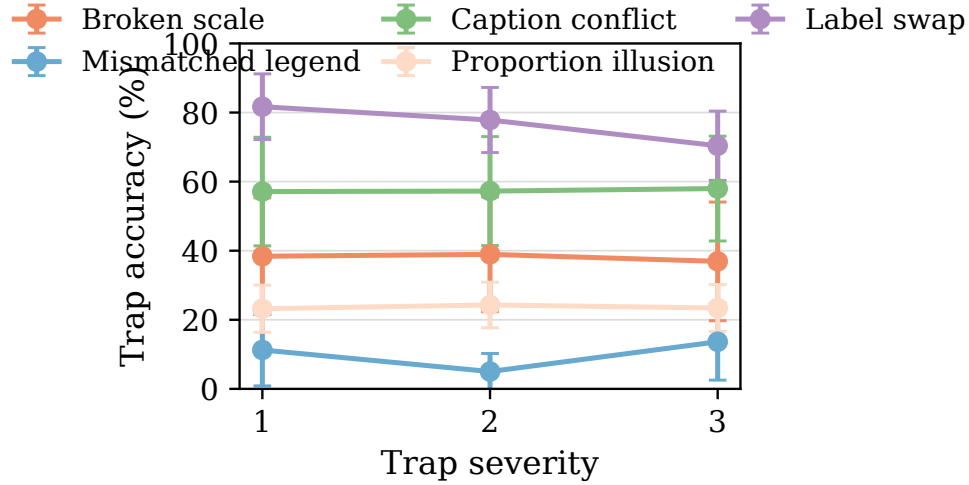


Figure 5: Severity effects are family-specific rather than uniformly monotonic: mismatched legend collapses to 5.0% trap accuracy at severity 2, while label swap remains at 70.4% even at severity 3.

The severity ladder is likewise an appendix diagnostic rather than a headline result. It records the family-specific pattern summarized in the discussion: mismatched legend collapses earliest, label swap degrades mostly at severity 3, and broken scale remains difficult at every level. Keeping the full plot in the appendix makes that non-monotonic structure reviewable without overloading the main results flow.

Appendix Figure 6 resolves one remaining diagnostic question for the reported open-weight slice. The completed rows show zero refusal for every model, which means the control-trap gaps reported in the main body are not artifacts of abstention behavior. We keep this plot in the appendix because it is a useful validity check rather than a headline benchmark claim.

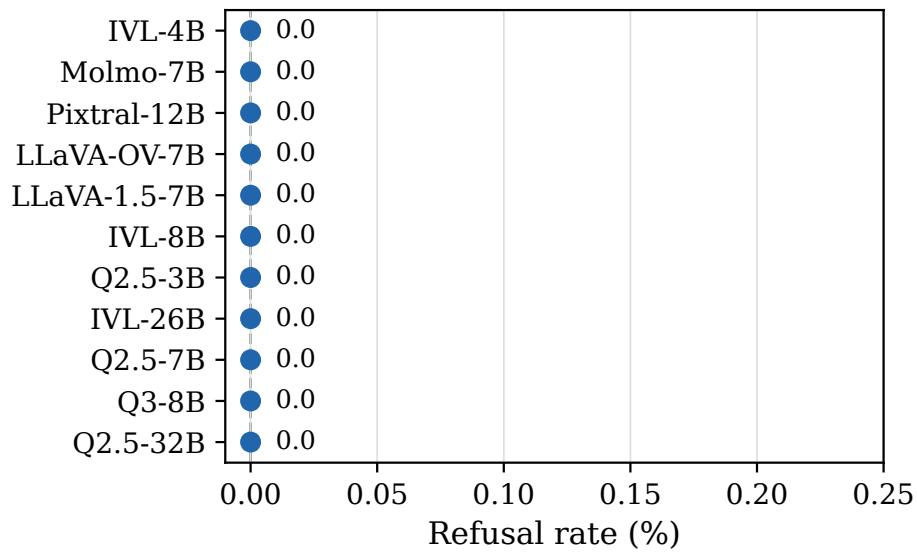


Figure 6: Refusal does not explain the completed open-weight slice: every model in the reported slice records 0.0% refusal over 6,000 evaluated responses, so the observed control-trap gaps come from answer changes rather than abstention.

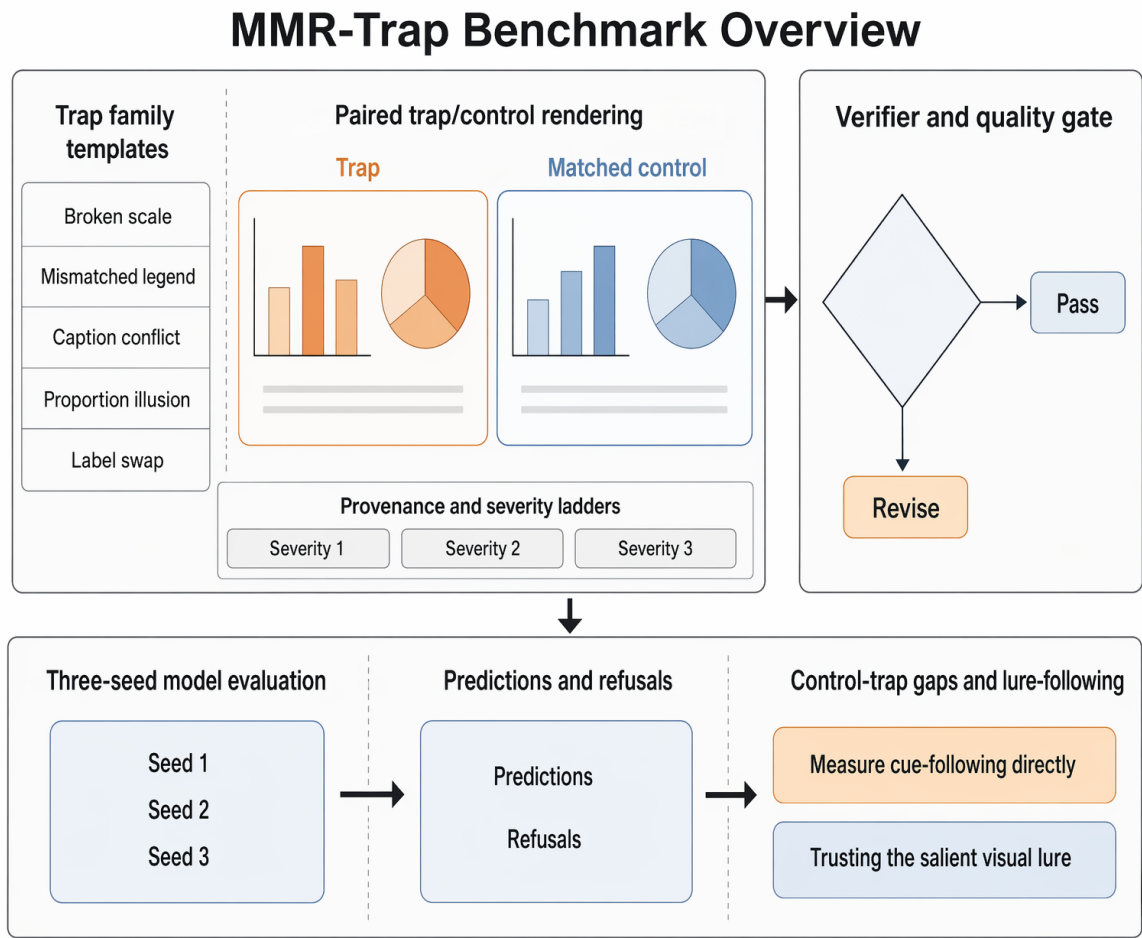


Figure 7: Appendix construction overview for the documented MMR-Trap slice. The figure keeps only the family bank, paired trap/control rendering, verifier plus severity ladder, and the three-seed evaluation path; the surrounding appendix text names the omitted provenance fields and replay details.

Appendix Figure 7 expands the body teaser into the documented construction contract while keeping only the stage names and one cue per stage inside the raster. The surrounding appendix text carries the provenance fields, verifier contract, and replay details that would otherwise force the figure back into microtext.

This overview also clarifies why the paper treats MMR-Trap as a contradiction-checking benchmark rather than a loose image collection. The construction ledger stores the template id, trap family, severity, control flag, verifier contract, and prediction traces needed to reproduce the control-trap gap analyses without depending on hidden local tooling.

The panel structure mirrors the study evidence bundle directly. The family template block corresponds to the benchmark specification tables, the paired rendering block corresponds to the saved trap/control item pairs, the verifier panel corresponds to the answer-extraction and quality-gate contract, and the evaluation panel corresponds to the three-seed scored rows summarized in the main results matrix. Keeping that mapping visible in the appendix lets a reader audit the benchmark as a reproducible research object instead of a black-box prompt collection.

The closing overview doubles as the paper’s compact replay interface: a reviewer can cross-check the documented templates, verifier behavior, severity fields, and per-model outputs without reconstructing the authoring workflow, which keeps the construction path behind the saved control-trap gaps visible.